

Concept activation vector in VQA setting

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Abstract

The vast research body in the domain of Artificial Intelligence does not only foster devising new concepts, but also combining the existing ones. The main aim of this paper is to apply Contrastive Activation Addition, a method of steering the behaviours of Large Language Models, to the setup of Visual Question Answering and study its effects. A Vision Language Model is utilised for a selected experimental scenario. The results indicate the occurrence of behaviour shift yet leave room for further research.

1 Introduction

With the ongoing AI revolution and its explosive growth in scientific research, both real-life applications and ideas' combinations follow. One notable and well-established challenge, which can benefit from the recent advancements and profound transformers' achievements, is the task of Visual Question Answering (VQA). The currently dominating methods originating from language modelling provide a positive outlook on this problem. At the same time, utilising Large Language Models (LLMs) and related methods poses new problems and enables a whole new body of research. An idea of influencing LLMs is among them, with Contrastive Activation Addition (CAA) being particularly interesting.

The primary objective of this study is to apply CAA to a multi-modal VQA context and analyse the steering effect on a Vision Language Model (VLM).

In Section 2, we refer to the key works relating to the two base concepts; in Section 3 the experimental setup and implementation details are described; finally, we provide the results in Section 4 and summarise the study in Section 5, also outlining vital extensions in Section 6.

2 Related works

2.1 Contrastive Activation Addition

Contrastive Activation Addition is a method for steering language models by modifying their activations during forward passes, introduced by Panickssery et al. in 2024 [4]. It involves computing a "steering vector" by taking the average difference in residual stream activations between pairs of prompts, which

demonstrate a target behavior (e.g., accuracy, refusal, sycophancy) and its opposite.

The mechanism is meant to isolate the model’s internal representations which closely match the desired inclination. During the inference, the steering vector is added at all token positions following the user’s prompt. The extent to which the desired behaviour is amplified or damped is determined by setting a numerical multiplier. According to the study, CAA can successfully complement or substitute other techniques, such as parameter optimisation, system prompt design or fine-tuning in certain applications.

2.2 Visual Question Answering

Visual Question Answering is the task of providing natural language answers to open-ended questions based on images. Spanning diverse areas such as computer vision, natural language processing and knowledge representation, VQA can be described as an "AI-complete" challenge [1]. In spite of requiring multi-modal knowledge, the task may be solved by composing single-task methods. VQA datasets embrace complex real-world images (eg. MS COCO [3]), synthetic scenes (eg. CLEVR [2]) or task specific samples. A crucial aspect within solving VQA tasks is to avoid the "Clever Hans" effect - a phenomenon in which a model exploits dataset or language biases instead of genuinely inferring from a given sample.

3 Design

3.1 CAA

CAA vectors are generated and applied on a single model layer and involve its activations. For vector generation, the activations at the appropriate token positions (related to the positive and the negative behaviours) are retrieved and their differences averaged over the training set. A varying number of samples may be used - training subsets of size 10 and 100 are researched in this work.

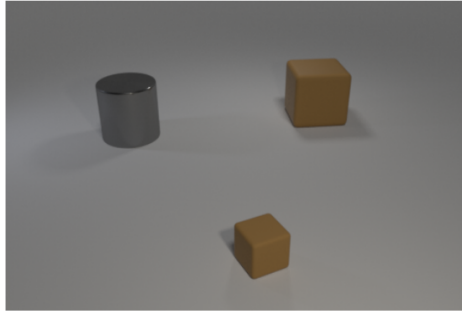
Vector application is realised by adding the steering vector at all token positions at the matching layer. CAA can be run with multipliers (which scale the steering vector and in effect the desired behaviour shift) - here ranging from -2 to +2 by 0.5.

3.2 Data

Researching CAA requires prompting a model with data in the form of questions and answers. In particular, two answers are necessary for one question - one is the true answer (reflecting the reality) and the other is the target answer (relevant to the researched problem). Thus the dataset applied must either provide both answers or allow generation of the target answer.

We use CLEVR in this study - a diagnostic VQA dataset, which provides entries as image-question-answer tuples, along with other extensive scene composition details [2]. CLEVR consists of 100,000 rendered 3D images featuring simple geometric shapes with varying attributes (size, color, material) and approximately one million algorithmically generated questions. The dataset comprises 90 natural language question families, obtained through matching predefined text templates with functional programs constituting images. Each image features approximately 10 questions. An example from CLEVR (with only some information from the sample) is depicted on Figure 1. Although the dataset itself provides only one answer per question, it is easily possible to complement it with desired target variants. Due to its synthetic nature, CLEVR could be also extended with new data by utilising the provided script or modifying it.

The approach of using existing data and appending target answers to it is employed in this study.



Q: There is a rubber	Q: What color is the
cube in front of the big	object that is on the
cylinder in front of the	left side of the small
big brown matte thing;	rubber thing?
what is its size?	A: gray
A: small	Q-type: query_color
Q-type: query_size	Size: 7
Size: 14	

Figure 1: An exemplary image from the CLEVR dataset along with questions, answers and program metadata [2].

3.3 Modelling

3.3.1 VLM

VQA can be realised with any model accepting image and text input and returning a text output. CAA in this setup, however, enforces the use of a language model. Hence, a Vision Language Model, in particular Google Gemma 3 IT (4B) is utilised in this experiment [5]. VLMs employ the transformer architecture, with the main components being a vision encoder, a projector and a 34 layer

deep LLM, which is the subject of CAA. Gemma 3 is based on the decoder-only architecture and accepts the vision modality, as well as provides a 128K-wide context window.

3.3.2 Target behaviour

We test CAA on a VQA setup which involves counting objects present in the image. A true answer (negative behaviour) remains the same as in CLEVR - a single number in range 0-10. A target answer (positive behaviour) is defined as a number randomly chosen from the valid range yet different from the true answer.

3.3.3 Contrastive pair construction

Input template A vital detail related to the utilised dataset and model is the construction of the contrast pairs and the evaluation examples. Positive and negative examples constituting the contrast pairs share prompts and differ in the answers appended at the end of the prompt. Specifically, the prompts for Gemma 3 IT are provided within a chat template and comprise: a BOS token, tokens indicating the start and end of a user’s message, a question, an image and a token indicating the start of the model’s output at the end. A system prompt may be prepended to the chat message when specified.

Samples used in assessment are fetched to the model just as chat template prompts, without the answers appended. Both variants are available in the dataset yet involved in the evaluation after the model responds.

Question structure Another design factor of crucial importance relates to the way of assessing the generated answers. In principle, a VLM may output a text response of any length and form, specifically providing the desired information in a sentence. On the other hand, some CAA setups and the use of CLEVR in particular entail working with a closed set of valid answers. Only these are to be taken into account to classify an answer as matching the target behaviour or its opposite. With the usual sentence generation the actual information (especially one of the valid answers) may be assigned relatively low activation and probability values for its output tokens, in favour of non-informative tokens which are to be placed in the output text (e.g., *Let’s, Here, the, answer*). Moreover, dealing with textual representations of a valid answer different from the defined one (e.g. numeric *4* vs. alphabetic *four*) poses a threat of obfuscating the study or hindering its reliability and consistency.

Therefore, in order to align the LLM mechanics with the evaluation scheme, a few subtle modifications to the data are introduced, similarly as in [4]. Firstly, the questions are provided as the actual text of a question concatenated with a multiple-choice list of possible options (valid answers) with answer letters. Secondly, the target and true answers are mapped to the option letters as well. Lastly, a system prompt is defined, forcing the model to respond with an option

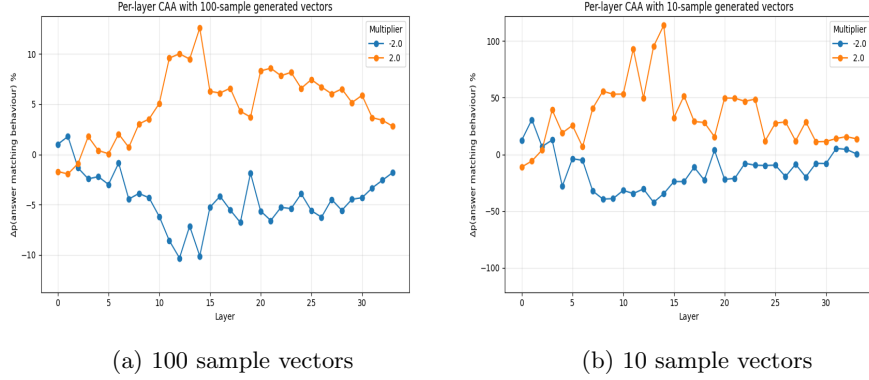


Figure 2: Relative shifts in target probabilities

letter. Thus only the tokens for the option letters representing the positive and negative answers are considered.

3.3.4 Evaluation

Similarly to Panickssery [4], the effect of adding a steering vector is measured by evaluating the model on a separate split of the dataset, scoping both true and target answer variants per question.

The steering mechanism is examined on a 100 sample subset of the CLEVR validation split featuring only the counting task questions. The probabilities of generating both answer variants are calculated from the last layer activations. At the same time, the probabilities normalised only to the target and true answer tokens are retrieved. Then, absolute and relative shifts (differences) from the no-steering evaluations are obtained. The results are collected separately for each vector layer, multiplier value and training size variant.

4 Results

The results feature the probabilities of the model generating the two answer variants for the studied behaviour in the counting task with a steering vector injected. First, the probability shifts are analysed per layer. Figure 2 consists of observations for one pair of multipliers (-2 and 2). Figure 2a shows the relative changes in the probability of returning an answer matching behaviour with respect to no steering. The chart presents the results for the related steering vectors applied to each of 34 model’s layers, having each vector generated from a sample of 100 examples. These results show that the steering into the defined behaviour is observed - both positive and negative steering significantly shift the target answer probability from the neutral state into the respective direction, with the maximum discrepancy in the relative behaviour shift at the level of 22.68 pp for layer 14.

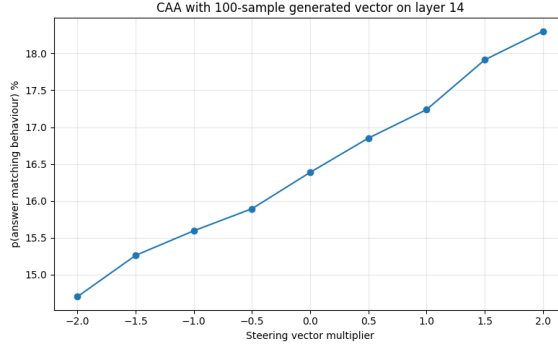


Figure 3: Steering effect by vector multiplier

According to the experiment, the steering effect is not uniform across the VLM’s language layers. Middle layers (7-27) are more prone to the behaviour influence, having on average 6.7 pp relative shift in the positive and 5.86 pp in the negative direction. First and last layers, though, are barely affected, achieving on average 2.36 pp and 2.4 pp respectively.

Similarly, the results of applying the small CAA vector are depicted in Figure 2b. This setup also allows for the maximum discrepancy on layer 14, despite the difference in the overall value range. Its average relative shifts are 48.72 pp and 23.15 pp for the middle layers (7-27) and 13.13 pp and 0.65 pp for the first and last layers (positive and negative steering respectively).

The relationships between the vector multiplier and the target answer probability is portrayed in Figure 3. The chart features results obtained for the 14th layer (where the greatest difference in probability shift from Figure 2 is observed) and shows a linear relationship.

5 Conclusion

This study indicates that CAA is applicable to VQA. The results of the layer-wise analysis (the two charts in Figure 2) comply with the characteristics of LLMs examined in the original CAA study [4] and suggest that both LLMs and VLMs expose an influence room in their hierarchical processing midst. The two charts in Figure 2 suggest that some steering may be realised also by vectors created with a small data sample, since both examined cases follow a similar pattern. The results for different multipliers shown in Figure 3 confirm that their value consistently determines the direction and extent of steering.

Although some clear patterns emerge from the results, it is necessary to outline key limitations and propose further investigation directions.

6 Future works

This work serves as an initial study, aimed at verifying the applicability of steering on a VQA setup. Its scope has been limited because of time constraints, excessive computational exploitation and elemental laboriousness. Hence, there are a few possibilities worth studying left.

Firstly, other tasks and behaviours may be examined. These embody tasks such as objects’ colour or size recognition, and steering into behaviours of, e.g. random or bimodal answering. Whether a steering effect may be exhibited in different setups and whether there are differences in their magnitudes should be verified.

In addition, the steering effect should be cross-examined on other data subsets - questions which represent tasks different from those used in the steering vectors’ creation. The steering may be also evaluated on the whole dataset or its sample featuring all question families.

Another relevant task would be to calculate the accuracy of the model’s answers and therefore assess the general capabilities while exposed to steering.

Surplus to this, the influence of the vector size (number of samples used to generate a steering vector) on the probability shift and the accuracy should be examined to determine whether there is a relationship or a sample size sufficient for achieving steering. Training samples of size eg. 10, 50, 100, 500 and 1,000 together with a fixed test set may be considered.

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